Video Transformers for Dynamic Breast Infrared Imaging Classification

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Abstract. Breast Cancer is a leading cause of mortality among women worldwide and early detection remains crucial for improving patient survival rates. Thermography offers a non-invasive imaging technique for identifying abnormal temperature patterns associated with malignancies. In this work, we propose a video transformer-based model for automatic BC classification from thermographic images using Universidade Federal Fluminense dynamic acquisition protocol. Our model achieved an Area Under the Receiver Operating Characteristic Curve score of 0.94 on a test set of 15 patients outperforming the results of the state-of-the-art set by 3D Convolutional Neural Networks.

Keywords: Breast cancer detection · Thermography · Transformers · Deep learning · Medical image analysis.

1 Introduction

Breast Cancer (BC) remains one of the main health challenges affecting women worldwide and it is the second most lethal form of cancer in Ecuador [4]. Since its introduction in 1960, Mammography (MG) has remained the gold standard for early detection [13,25], with screening programs demonstrating effectiveness in reducing mortality rates. However, MG faces limitations: reduced efficacy in younger women with dense breast tissue, ionizing radiation risks, and reliance on manual interpretation of low-contrast lesions with varied morphologies [18,23]. In contrast, Thermography (TG) offers a radiation-free, non-invasive, and cost-effective alternative for BC screening that is applicable to all age groups without contraindications. Moreover, when combined with machine learning techniques, TG demonstrates significant potential for early BC detection [19].

Recent advances in Artificial Intelligence (AI), particularly in Deep Learning (DL), have significantly transformed computer-aided detection (CAD) systems for medical imaging [17]. Although convolutional neural networks (CNNs) currently dominate image processing, emerging transformer-based models demonstrate superior capabilities in global context modeling [8]. This study investigates the application of transformer architectures, specifically Vision Transformers (ViTs), for automated classification of BC using the UFF's Dynamic Infrared Thermography (DIT) protocol, which employs a sequence of temperature maps per patient [27]. This Infrared (IR) image acquisition method established the superiority of dynamic protocols for highlighting vascular behavior [22]. In this work we hypothesize that ViTs can better model temporal dependencies in thermal sequences compared to CNNs. As a consequence, we examine both video and image-based ViTs implementations and compare their performance with that of CNNs-based architectures. Our primary objective is to improve diagnostic accuracy in BC classification through TG imaging. . The following are the proposed research questions for this research:

1. **RQ1:** Do ViTs outperform CNNs in BC classification from thermographic images due to their superior ability to model global contextual relationships?
2. **RQ2:** Does transfer learning with pre-trained ViTs architectures improve classification accuracy compared to models trained from scratch for this medical imaging task?
3. **RQ3:** Does combining pre-trained ViTs models with dynamic acquisition protocols enhance diagnostic classification performance for breast cancer detection in thermogram images compared to conventional approaches?

In this work, we fine tune a Video Vision Transformer (ViViT) to classify thermal images under the DIT protocol. Our study makes the following key contributions: (1) We extend Vision Transformer (ViT)-based approaches to DIT protocol, leveraging sequential image analysis to improve BC classification performance using a patient level dataset. (2) We evaluate the impact of Fine-Tuning (FT) using pre-trained ViTs. (3) We conduct experiments on two different versions of the DMR-IR dataset (thermal map and thermal image) and compare performance to a simple CNN architecture. (4) To the best of our knowledge, this is the first study accounting the use of Video Vision Transformers (ViViTs) in DIT protocol for TG in BC classification. Our model achieved an Area Under the Receiver Operating Characteristic Curve (AUC)-score of 0.9350 ± 0.0094 in 10 fold cross validation setup and 0.9444 on a separated test set of 15 patients with stratified labels. This work advances the state-of-the-art set by [6], where 3D CNNs achieved 93.51% AUC in binary classification, and compliments the work by [11], which achieved 95.7% in Static Infrared Thermography (SIT) protocol [22,27].

The organization of this paper is presented as follows: In Section 2, a brief introduction to main characteristics about the TG protocols for screening BC is presented. Section 4 presents the related works to our research questions. Our proposed methodology is presented in Section 5. Results are presented in Section 6. Finally, discussion over main findings is presented in Section 7 and this work's conclusions in Section 8.

2 Advantages of Dynamic Infrared Thermography in Breast Cancer

By acquiring the body's natural emissions of infrared radiation to map the body's temperature distribution, TG is a non-invasive imaging modality that facilitates BC detection [13,19,25]. Due to the heightened metabolic activity of cancerous cells, there is an increased blood flow to the tumor, which promotes the development of new blood vessels presenting abnormal behavior under temperature modifications. These anomalies typically appear as thermal asymmetries between the contralateral sides in the breast region under induced thermal stress.

Table 1: Comparison of MG and TG

Parameter	**Mammography**	**Thermography**
Basis	X-rays	Infrared radiation
Device	X-ray machine[3]	Infrared camera
Output	Anatomical image	Thermal map
Abnormality Type	Anatomical	Physiological
Equipment cost	≈ \$80K[a]	≈ \$0.4K[b]
Age	over 40 years	No age barrier
Women with implants	Not so effective	Effective
Breast Density	False Positives	Not affected
Further Investigation	Unnecessary biopsies	Fewer biopsies
Pain and Fear	Breast Compression	No touch
Radiation	Ionizing radiation	None
FDA approval	Cancer diagnosis	Adjunct to MG

[a] https://www.blockimaging.com/
[b] https://www.flir.com.mx

Moreover, unlike MG, which is less effective in younger patients due to the characteristics of denser breast tissue, TG can be safely applied across all age groups and is generally more cost-effective than MG and other imaging modalities. It can also be applied to females with dense breast tissue, women who are pregnant or breast feeding. Although the FDA approved TG as a cancer risk assessment tool in 1982, it is typically used as an adjunct to MG rather than a standalone diagnostic method [13,20]. Advances in thermal sensing equipment, image processing, machine learning and AI pipelines have contributed to more feasible and accurate CAD systems for infrared TG.

TG images can be obtained by using two different protocols: SIT and DIT. The former consists in capturing the thermal images under controlled conditions of frontal, oblique and lateral breast views [20]. The latter measures the thermal response to a thermal stress applied to the patient [20,25]. The UFF's DIT protocol captures an image every 15 seconds over an interval of 5 minutes, resulting in a sequence of 20 images per patient [6,27]. Despite its advantages, TG is susceptible to procedural inconsistencies and thermal noise interference. A major challenge lies in the non-specificity of thermal anomalies, which complicates accurate interpretation. For instance, elevated temperatures may result from non-cancerous conditions such as infections or inflammations. A comparative overview of TG and MG is presented in Table 1.

3 Approaches for Sequence Modeling

3.1 Image Vision Transformers

In 2017, the Transformer architecture was introduced in the seminal paper "Attention Is All You Need" as the first sequence transduction model relying entirely on attention mechanisms [26]. This architecture replaced recurrent layers with an encoder-decoder structure based on multi-head self-attention, establishing itself as the state-of-the-art approach for Natural Language Processing (NLP) tasks.

Attention mechanisms had been previously explored in image processing [8]. However, this work demonstrated that a pure Transformer architecture could be directly applied to sequences of image patches, treating them analogously to tokens in NLP tasks with minimal modifications to the original architecture. An image patch, in a ViT is a fundamental hyperparameter that determines how an input image is divided into smaller segments (patches) before being processed by the transformer architecture. The patch size defines the dimensions of each non-overlapping block that the image is split into. For an input image of size $H \times W \times C$ (Height $\times$ Width $\times$ Channels) and a patch size of $P \times P$, the number of patches is $N = \frac{H}{P} \times \frac{W}{P}$; where each patch becomes a token of dimension $P \times P \times C$. For instance, if the image is $(224 \times 224 \times 3)$, $N = \frac{224}{16} \times \frac{224}{16} = 14 \times 14 = 196$ tokens.

Dosovitskiy et al. showed that ViTs achieve superior performance when pre-trained on large-scale datasets such as ImageNet-21K and JFT-300M, followed by FT for downstream tasks, outperforming state-of-the-art CNNs. Additionally, their work provides key implementation insights: (1) **Positional embeddings** show no significant improvement when extended to 2D representations; (2) **FT** employs higher-resolution images while maintaining original patch size [15]; (3) **Transfer Learning (TL)** replaces the classification head with a zero-initialized linear layer matching the target output dimension; (4) **Training** requires strong regularization for scratch training on ImageNet with cosine learning rate decay; (5) **Preprocessing** adheres to guidelines from [15].

3.2 Video Transformers

Video can be understood as a sequence of images through a time dimension that introduces motion and deformations on them [21]. Because of this sequential nature present in video information, it would be expected that ViTs to be effective in video modeling [3]. The *TimeSformer* stands as the first convolution-free architecture that adapts the ViT architecture proposed in [8] to video by extending the self-attention mechanism from the image space to the space-time 3D volum [3]. Their main idea is to treat video as a sequence of patches extracted from the individual frames, where each patch is linearly mapped into an embedding space with positional information [3].

There is active research on ViViTs, with several notable models proposed in recent literature (e.g., ViViT [1], MViT [10]). Most improvements across different architectures primarily address the computational complexity inherent in

Transformer models, which scales quadratically with respect to the sequence length T (i.e., $\mathcal{O}(T^2)$). Additionally, these approaches tackle challenges related to spatial redundancy, temporal fidelity, and inductive biases [21].

3.3 CNN-based Sequence Modeling

While Recurrent Neural Networks (RNN) architectures (e.g., LSTM and GRU) have traditionally been the default choice for sequence modeling, recent studies suggest that CNNs may outperform them in certain tasks [2]. The Temporal Convolutional Network (TCN) is a general-purpose convolutional architecture for sequence prediction characterized with two key properties: (1) *causal convolutions*, which prevent information leakage from future to past data, and (2) a *fixed output length*, enabling the model to map variable-length inputs to outputs of the same length [2].

A significant advantage of TCN is its inherent parallelism, where predictions for later time steps do not depend sequentially on earlier ones. This is achieved by processing the same filter across each layer, allowing the model to handle long input sequences efficiently during both training and evaluation.

In contrast, Keras [5], a popular DL framework, provides a Time Distributed Layers (TDL) wrapper that applies a given layer to every temporal slice of an input. By reusing the same weights across all timestamps in the 3D input tensor, this approach ensures parameter efficiency. Recent work by [7] demonstrates that combining TDL with LSTM can simultaneously improve feature extraction and temporal pattern recognition in sequential data, leading to enhanced accuracy in Human Activity Recognition (HAR) tasks.

4 Literature Review

This section provides a systematic literature review following the methodology proposed in [14] to find relevant works in the area of this research goals and questions. To ensure comprehensive coverage of ViTs and TG applications, we employed the Population, Intervention, Comparison, Outcome, Context (PICOC) framework to engineer the search string. We focused specifically on studies that combined thermographic imaging with advanced computer vision techniques, particularly ViTs, in the context of BC detection.

4.1 Search Methodology

We conducted targeted searches across four major digital libraries, such as: ACM Digital Library, IEEE Xplore, Scopus, and Springer Link. Additional repositories were examined but yielded no relevant results aligned with our research scope. We employed the following Boolean search query:

```
(thermogram OR thermography) AND
(vision transformer OR vit) AND
(cnn OR convolutional neural networks OR deep learning OR machine learning) AND
(improve OR enhance) AND
(classification OR detection OR diagnosis OR segmentation) AND
breast cancer
```

The search query was designed to identify research papers comparing ViTs and CNNs models. Given the scope of our study, we adopted broad inclusion criteria for DL-based computer vision models applicable to TG image classification. We systematically excluded studies that did not primarily focus on TG as the screening modality, such as those employing ultrasound, magnetic resonance imaging, or MG. Furthermore, non-specific sources (e.g., journals containing multiple unrelated articles) were discarded. During the search for BC research, we also encountered numerous studies on histopathological and genetic analyses, which were outside our scope and thus excluded.

The execution of the proposed search strategy yielded 24 primary studies after removing duplicates. Subsequently, each study underwent a screening process where titles, keywords, and abstracts were evaluated against the predefined inclusion and exclusion criteria. This selection process resulted in 4 relevant publications for the literature review. The scarcity of publications investigating ViT and TG suggests significant opportunities for novel contributions, with only one study [11] addressing research objectives comparable to ours.

4.2 Deep Learning for Dynamic Breast Thermography

According to our proposed search strategy for literature review, the first study employing ViTs for BC classification using TG images is reported in [11]. Their approach utilizes the standard ViT-Base architecture with 12 layers, 12 attention heads, and a 768-dimensional embedding spac [8]. Similar to our work, their methodology leverages the transformer's self-attention mechanism for TG image classification into *Cancerous* or *Healthy* categories. The authors investigate the impact of patch size on classification accuracy, demonstrating that 32×32 patches yield superior performance compared to the standard 16×16 configuration. Also, it is worth to mention that their implementation does not incorporate pre-trained weights on natural images or data augmentation techniques. Their approach achieved an AUC of 95.7% for binary classification on the DMR-IR datase [22], using a total of 950 samples with an analysis unit per TG image. However, the study was limited to SIT acquisition protocol.

To benchmark our methodology against CNNs-based approaches for DIT, we compare with [6], which employs 3D-CNNs to recognize patterns changes in a sequence of TG images. Cancer identification using a sequence of images is similar to action recognition and depend on spatio-temporal models that extract space-time feature information from data. While 2D CNNs applied to individual frames remain competitive for action recognition [24], 3D convolutions inherently capture motion information across adjacent frames [12,16]. In [6], each patient's data is represented as a hypercube constructed by stacking 20 preprocessed frames. The preprocessing stage involved the segmentation of the region of interest from the original TG images and their resized to $32 \times 32 \times 3$. Baffa et al. enhanced the model generalization through data augmentation with the following operations: rotations up to $15°$, horizontal flips, shearing transformations and adding noise. According to them, 3D CNN is capable of considering the spatial relationships in each image and their evolution over a temporal dimension.

Their experiments employed the DMR-IR dataset, comprising 2 980 images from 149 patients. Under a K-fold cross-validation strategy, their model achieved an AUC score of $93.51\% \pm 4.03\%$.

Despite the current trend of researching modern DL architectures for BC classification in infrared TG, it should be noted that texture-based CAD systems can accurately classify BC in TG, achieving over 90% accuracy, due to the rich texture content present in the images [20]. A significant advantage of these texture-based methods is their computational efficiency, particularly when dealing with limited datasets, as they require substantially fewer resources than their DL counterparts. This efficiency, however, comes at the cost of requiring manual feature engineering, whereas DL approaches automate the feature learning process.

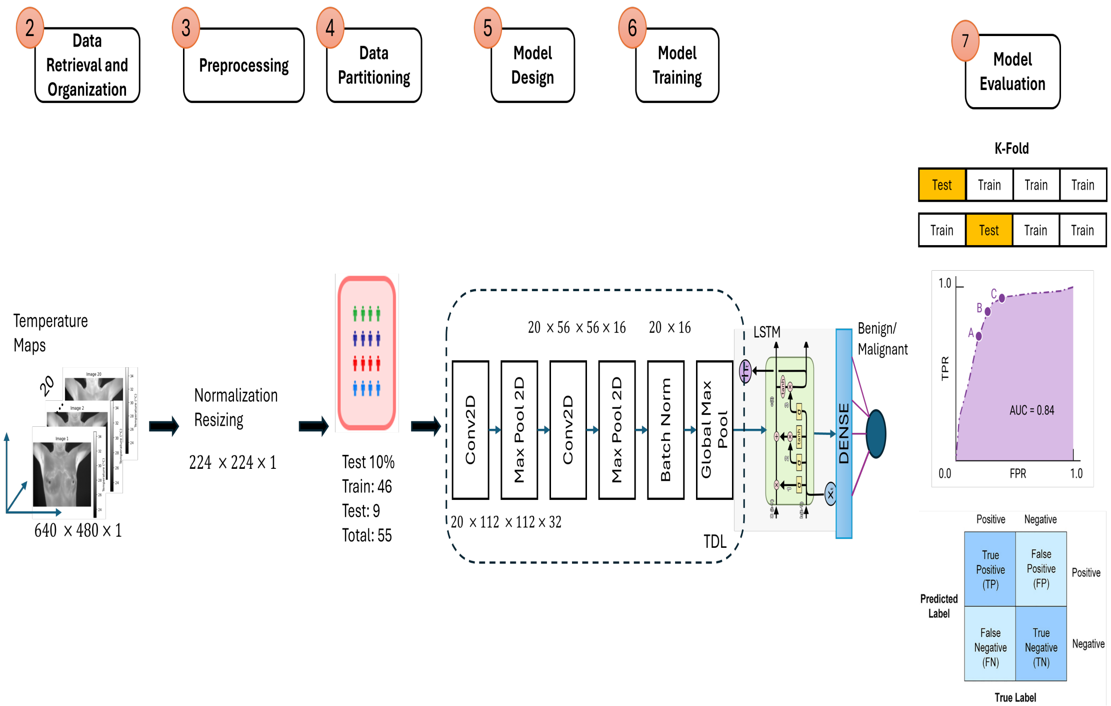
5 Methodology

This section outlines the methodological procedures employed to develop and validate our set of models for BC using the DIT protocol, specially the efficacy of ViTs models. The most similar work to our approach was proposed in [11], where the authors trained a ViT model from scratch using the same dataset but the SIT protocol. Their study does not evaluate the performance of pre-trained ViTs, which is studied now here.

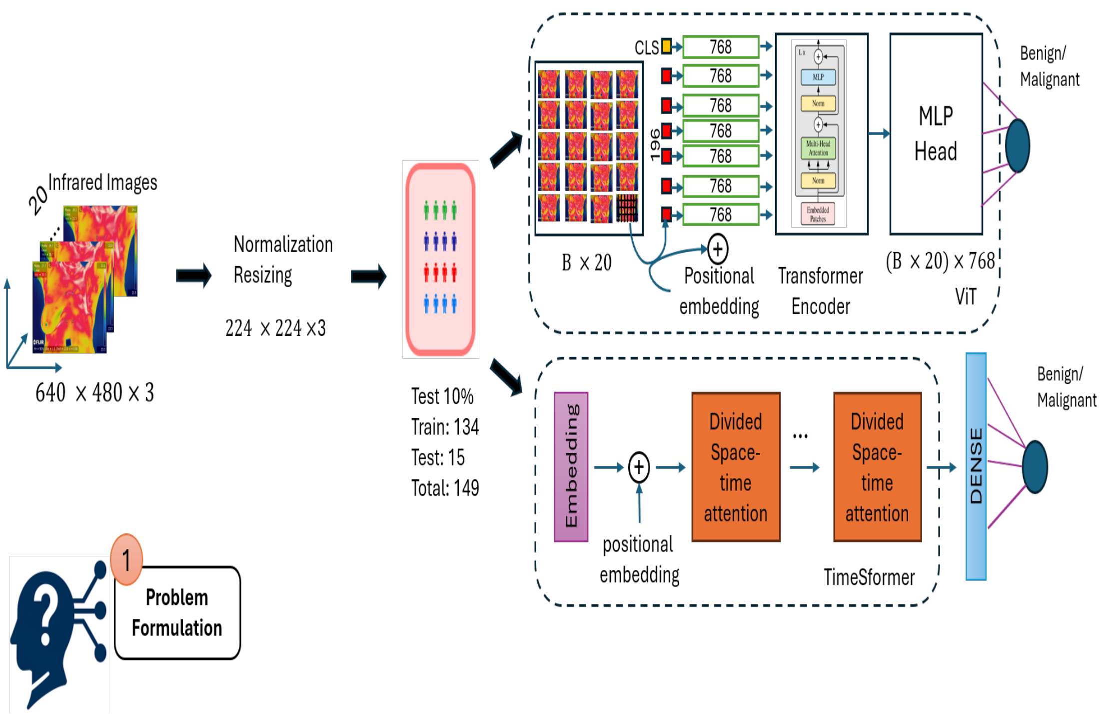
From our literature review there is no previous automated classification of BC by using DIT protocol and ViViTs. To benchmark our results, we consider the work of Baffa et al [6], where 3D CNNs are employed to study the evolution of the sequence of TG images.

The methodology consists of seven main stages (Figure 1), where some steps vary depending on the specific experiment. First, we formulated the machine learning problem. Thereafter, we retrieved the sequences of Temperature Maps (TM) ($\mathbb{D}_{\mathrm{TM}} \in \mathbb{R}^{640\times480\times1}$) and IR images ($\mathbb{D}_{\mathrm{IR}} \in \mathbb{Z}^{640\times480\times3}$) for each subject from the DMR-IR dataset, creating two distinct data subsets. The TM dataset comprises temperature matrices with a precision of up to two decimal places and an accuracy of 0.02°C, whereas the IR dataset consists of three-channel RGB infrared images with pixel values ranging from 0 to 255.

Next, we set the preprocessing pipelines for each dataset according to its specific characteristics, including data augmentation during training. Afterward, for $\mathbb{D}_{\mathrm{IR}}$, we applied a stratified holdout sampling technique, reserving 10% of the cases as a test set. We then designed the models using a 5-dimensional input tensor $\mathcal{X} \in \mathbb{R}^{B\times F\times C\times H\times W}$, where: B denotes the batch size, F represents the number of frames, C corresponds to the number of channels, and H and W indicate the height and width of the TM or IR images, respectively. Due to the small dataset size, we evaluated the models using K-fold cross-validation. Finally, we selected the best-performing model and evaluated it on the test set. We employed AdamW as the optimizer together with checkpoint and early stopping callbacks. All experiments were performed using Kaggle's GPU 1 x Tesla P100 of 3584 CUDA cores, 16 GB GDDR6 and up to 30 GB of RAM.



(a) Methodology for Temperature Maps: depicts the CNN network with TDL



(b) Methodology for Infrared Thermography Images: depicts the ViT and ViViT models

Fig. 1: Overview of the methodology used in this research. It comprises seven stages: (1) Problem Formulation, (2) Data Organization, (3) Preprocessing, (4) Data Patitioning, (5) Model Design, (6) Model Training, and (7) Model Evaluation for (a) Temperature Maps and (b) Infrared Thermography Images.

5.1 Problem Formulation

Our objective is to train a model to classify a sequence of TG data into either *Benign* or *Cancer*. Consequently, we want to find a mapping function $f_\theta : \mathbb{R}^{F\times C\times H\times W} \to [0,1]$ with parameters θ, where the predicted class probabilities for an input sequence $\mathbf{x} \in \mathcal{X} \subset \mathbb{R}^{F\times C\times H\times W}$:

$$P(y=1 \mid \mathbf{x}) = f_\theta(\mathbf{x}) \quad (\text{for } \mathbf{x} \in \mathcal{X}) \tag{1}$$

Considering that a video classification aims to predict the class of a given input sequence of frames [21], the final prediction $\hat{y} \in \{0,1\}$ is:

$$\hat{y} = \mathbb{I}\left(P(y=1 \mid \mathbf{x}) \geq 0.5\right) \tag{2}$$

The optimization minimizes the binary cross-entropy loss:

$$\mathcal{L}(\theta) = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right] \tag{3}$$

where $y \in \{0,1\}$ are the ground truth labels.

In this context, we hypothesize that ViViTs can outperform 3D CNNs in BC classification for TG.

5.2 Model Fine-Tuning

FT is a TL technique where the weights of a pre-trained model are further trained on a smaller, task-specific dataset. Typically, the classification head is replaced to adapt the model to the nuances of the new task. This process can update all layers of the pre-trained model or gradually unfreeze some of them (layer-wise fine-tuning). We adopt the formal definitions for TL and FT given in [9]:

Definition 1 (Transfer Learning). *Given an original domain $\mathcal{D}^S$ with an original learning task $\mathcal{T}^S$, a target domain $\mathcal{D}^T$ with a target task $\mathcal{T}^T$, the transfer learning operation seeks to improve the learning of a prediction function $\phi^{\mathrm{T}}(\cdot)$ in $\mathcal{D}^T$ using the knowledge acquired in $\phi^{\mathrm{S}}(\cdot)$ in $\mathcal{T}^S$, through ϕ^S; where $\mathcal{D}^S \neq \mathcal{D}^T$ or $\mathcal{T}^S \neq \mathcal{T}^T$.*

Equations (4) and (5) represent the transfer learning operation and model prediction, respectively. Here, L denotes the number of layers in the pre-trained model, and $\mathcal{A}$ refers to a set of operations that replace the original classification head.

$$\mathbb{T}_L\left\langle \phi^{\mathrm{S}}(\mathcal{D}^S), \mathcal{D}^T \right\rangle = \mathcal{A}\left(\phi^{\mathrm{S}}(\mathcal{D}^T)\Big|_0^L\right) \tag{4}$$

$$\mathcal{Y}^T = \mathbb{T}_L\left\langle \phi^{\mathrm{S}}(\mathcal{D}^S), \mathcal{D}^T \right\rangle = \phi^{\mathrm{T}}(\mathcal{D}^T) \tag{5}$$

Definition 2 (Fine-Tuning). *Given an original domain $\mathcal{D}^S$ with an original learning task $\mathcal{T}^S$, a target domain $\mathcal{D}^T$ with a target task $\mathcal{T}^T$, the fine-tuning procedure enhances the learning of a target function $\mathcal{Y}^T = \phi^{\mathrm{T}}(\mathcal{I}^T)$ by retraining r layers of the original model $\mathcal{Y}^S = \phi^{\mathrm{S}}(\mathcal{I}^S)$ with L layers, where $\gamma = L - r$.*

Equations (6) and (7) represent the fine-tuning operation and the fine-tuned model prediction, respectively. Full model update occurs when $r = L$. The operation $\mathcal{A}$ involves replacing the original classification head of the pre-trained model. For instance, in the case of TimeSformer, it would replace the original 400-class output for Kinetics-400 with a new head appropriate for the target task.

$$\mathbb{F}_T \left\langle \phi^{\mathrm{S}}(\mathcal{D}^S), \mathcal{D}^T \right\rangle = \mathcal{A}\left(\phi^{\mathrm{S}} \left(\phi^{\mathrm{S}}(\mathcal{D}^T)\Big|_0^{\gamma-1} \right)\Big|_\gamma^L \right) \tag{6}$$

$$\mathbb{F}_T \left\langle \phi^{\mathrm{S}}(\mathcal{D}^S), \mathcal{D}^T \right\rangle = \mathcal{A}\left(\phi^{\mathrm{S}}(\mathcal{D}^T) \right) \tag{7}$$

5.3 Data Collection, Preprocessing and Data Augmentation

DMR-IR dataset contains TG data of BC patients acquired under either the SIT or the DIT protocol. This data is provided in two formats: TM and IR images. The former comprises a single–channel, two–dimensional matrix of floating-point temperature values captured by the FLIR SC620 infrared camera with a measurement accuracy of $\pm 2\,^{\circ}\mathrm{C}$ [6]. The latter consists of RGB images acquired from the same camera. In both cases, the resolution is 640×480 pixels. However, as noted in [11], the number of subjects and images varies across studies, making direct comparisons difficult. Table 2 summarizes the characteristics of $\mathbb{D}_{\mathrm{TM}}$ and $\mathbb{D}^{\mathrm{IR}}$.

Even though all images in $\mathbb{D}_{\mathrm{IR}}$, are `.jpg` files, we found that 28 cases presented images of 160×120 and 1 case of 800×600. All of which were resized to 640×480. Also, a total of 21 images were copies of a previous sequence. This happened to 7 cases.

Table 2: DMR-IR dataset characteristics

Feature	$\mathbb{D}_{\mathrm{TM}}$	$\mathbb{D}_{\mathrm{IR}}$
Type	Temperature Map	Infrared RGB Image
Matrix size	640×480	640×480
Channels	1	3
Number type	float(temperature)	integer(pixel)
Total patients	55	149
- With cancer	36	87
- Without cancer	19	62
Frames per patient	20	20
Total images	1 100	2 980
Training set patients	46	134
Test set patients	9	15

For $\mathbb{D}_{\mathrm{TM}}$, normalization was performed using the entire patient sequence. As for the case of $\mathbb{D}_{\mathrm{IR}}$, predefined normalization values were applied to match the requirements of the pre-trained networks employed. Since the DIT protocol operates at the patient level, the number of available samples is inherently limited. To mitigate this constraint, data augmentation was applied exclusively to the training fold. The augmentation strategy differs from that of [6], incorporating only the following transformations: horizontal flip, rotation up to 15°, and resize. Other augmentation techniques were excluded to preserve the thermal properties of the data, which differ significantly from natural images. Building upon this idea, we do not consider of use to apply shearing transformations and introducing noise.

5.4 Model Development

Two categories of models were studied: (1) spatio-temporal architectures, and (2) spatial-only. The former experiments with TDL and ViViTs. While, the latter, applies the basic ViT designed to the flatten inputs (i.e. $(B \times F, C, H)$) without temporal modeling. Both PyTorch and Keras/Tensorflow were used as the deep learning backends. Table 3 presents the models studied, along with the source domain dataset used when FT was applied for model adaptation. The input tensor dimensions, deep learning backend, and number of parameters are also provided.

Time Distributed Layer Model: Building upon the hybrid architecture proposed in [7], we implement a sequential model combining TDL for spatial feature extraction and Long Short Memory (LSTM) for temporal modeling. Each frame undergoes spatial feature extraction through resizing to 224×224 resolution, processing through two convolutional blocks (32 and 16 filters with 3×3 kernels and ReLU activation), max pooling (2×2) after each convolution, and batch normalization for stable training. Subsequently, temporal processing is performed via global max pooling to reduce spatial dimensions to 16 features per frame, followed by an LSTM layer (16 units) with ℓ_2 regularization ($\lambda = 0.01$) to model temporal dynamics. Finally, the classification head consists of a dense layer (16 units, ReLU) with ℓ_2 regularization, dropout ($p = 0.5$) for additional regularization, and a sigmoid output for binary classification.

The TDL wrapper employs weight sharing across all frames, applying identical transformations to each temporal slice. This architecture efficiently integrates spatial feature learning through convolutional operations, temporal modeling via recurrent connections, and regularization techniques (ℓ_2 and dropout) to mitigate overfitting. During training, data augmentation was applied, and early stopping with model checkpointing was employed based on validation loss. This model was engineered to account for the lack of patients in the $\mathbb{D}_{\mathrm{TM}}$ dataset. It has a total of 7 377 trainable parameters and was coded using Keras/Tensorflow. Resizing of TM was carried out due to memory constrains.

Basic Vision Transformer: ViTs are inherently spatial architectures. We designed our ViT-based classifier to process complete sequences of 20 thermal frames per case, maintaining compatibility with the standard 5D input tensor format (B, F, C, H, W). We construct our model from the base architecture proposed in [8].

The architecture comprises three principal components: (1) a pre-trained ViT-B/16 backbone, from which the classification head has been removed; (2) frame-wise feature extraction where input sequences are reshaped from (B, F, C, H, W) to $(B \times F, C, H, W)$, processing each frame independently through the ViT encoder and extracting features vectors from the final layer with $D = 768$; (3) a linear classification head that projects these features to logits of shape $(B \times F, 1)$, and computes temporal mean predictions $(B, 1)$.

This design enables parallel processing of all frames while preserving sequence information through temporal averaging. The model accepts the same 5D tensor format as traditional video architectures but leverages ViT's superior spatial modeling capabilities for thermal image analysis. A total of 85 799 425 parameters are to be trained, when using 16×16 patches and 87 456 001, when the size of the patches is increased to 32×32. We experimented the ViT with initialized ImageNet1K weights and from scratch for the latter case.

Video Transformer Since video classification requires both spatial and temporal processing of image sequences, we implement a TimeSformer-based architecture [3] for DITanalysis. Generally, models for video data need to address two key challenges: (1) joint spatio-temporal feature learning, and (2) handling variable-length sequences. However, in DIT, this second challenge is mitigated by the standardized protocol where each patient examination contains exactly 20 IR frames.

Our TimeSformerThermoClassifier model comprises a pre-trained TimeSformer-base model, which was finetuned on Kinetics-400, and a modified classification head. The TimeSformer model is configured to process an input of $F = 8$ frames at $H \times W$ resolution; it employs spatio-temporal attention across divided space-time patches and maintains a hidden dimension of $D = 768$ across all layers. For the classification head, the original Kinetics-400 classification layer is replaced with a single-output linear layer to perform binary prediction; a sigmoid activation is implicitly used through the binary cross-entropy loss.

The forward pass processes input tensors $\mathbf{X} \in \mathbb{R}^{B \times F \times C \times H \times W}$ through a sequence of operations: first, frame rearrangement is applied when needed to match TimeSformer's expected (B, C, F, H, W) format; subsequently, spatio-temporal feature extraction is performed via patch embedding (typically 16×16) and divided space-time attention layers; following this, global average pooling of the spatio-temporal features is conducted; and finally, binary classification is achieved through the final linear layer.

This architecture leverages TimeSformer's proven effectiveness in video understanding while adapting it specifically for thermal medical imaging, where temporal consistency is guaranteed by the DIT acquisition protocol. However, the base architecture used in this research is able to hold up to 8 frames. To account to this fact we developed a random sampling from sequence code which allowed to extract up to 8 frames from the 20 without altering the order of the sequence.

Table 3: Specifications of the source domains (pre-training datasets), target domain (fine-tuning dataset), input tensor, framework, and parameters for each model variant tested.

Model	**Pre-Trained**	**Source Domain** $\mathcal{D}^S$	**Target Domain** $\mathcal{D}^T$	**Input Tensor**	**Framework**	**# Parameters**
ViT(16×16)	Yes	ImageNet-1K	$\mathbb{D}_{\text{IR}}$	$(B \times 20, 3, 224, 224)$	PyTorch	85 799 425
ViT(32×32)	Yes	ImageNet-1K	$\mathbb{D}_{\text{IR}}$	$(B \times 20, 3, 224, 224)$	PyTorch	87 456 001
ViT(32×32)	No	None	$\mathbb{D}_{\text{IR}}$	$(B \times 20, 3, 224, 224)$	PyTorch	87 456 001
ViViT / TimeSformer	Yes	Kinetics-400	$\mathbb{D}_{\text{IR}}$	$(B, 8, 3, 224, 224)$	PyTorch	121 259 521
TDL + LSTM	No	None	$\mathbb{D}_{\text{TM}}$	$(B, 20, 1, 224, 224)$	TensorFlow	7 377

5.5 Experimental Protocol

Figure 2 presents a guide for the experiments carried out in this research and compares the performance of the different models on the *F1-score* metric for the *K-Fold Cross Validation* strategy. Sequential models are highlighted.

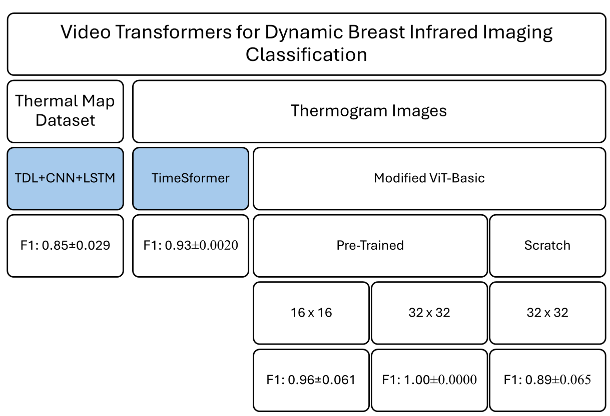


Fig. 2: A comparison of F1-score performance of all the Models for BC in DIT in this research. Highlighted are the sequential models like TDL and ViViT

Our experimental protocol varied across modalities and architectures. For TM data, we employed 3-fold cross-validation trained for 50 epochs using AdamW

Table 4: Training protocols for different modalities and architectures

Modality	Model	Training Configuration	Validation	Epochs
TM	TDL+CNN+LSTM	AdamW, $k = 3$ CV	Loss	50
IR	ViT (16×16) pre-trained	AdamW, lr=3e-5, $k = 10$ CV	F1-score	20
	ViT (32×32) pre-trained	AdamW, lr=3e-5, $k = 10$ CV	F1-score	20
IR	ViT (32×32) scratch	AdamW, lr=1e-4, wd=0.05 LR warmup + cosine (min lr=1e-6), $k = 10$ CV	Loss	50
IR	ViViT	AdamW, lr=3e-5, $k = 10$ CV	F1-score	20

optimizer while monitoring validation loss. For IR images with pre-trained ViT models (both 16×16 and 32×32 patch sizes), we used 10-fold cross-validation for 20 epochs with a learning rate (lr) of 3e-5, optimizing for validation F1-score. However, when training the 32×32 patch ViT from scratch, we extended training to 50 epochs with higher initial lr (1e-4), weight decay (0.05), and implemented lr warmup with cosine annealing (minimum lr=1e-6), while monitoring validation loss. The ViViT model followed similar configuration to pre-trained ViTs (20 epochs, lr=3e-5) monitored by F1-score. Table 4 presents the training protocols used for the different modality-architecture combinations.

6 Results

In this section, we discuss the quantitative performance of the different models in our proposed experiments (Table 5) and compare our work with the most relevant studies identified through our literature review (Table 6).

Our work is similar to that of Garia et al. [11] in the use of Transformer-based models for breast cancer classification using Thermography (TG). However, we address dynamic infrared thermography (DIT), which requires a different methodological focus. Specifically, we investigate: (1) a temporal deep learning (TDL) approach using Time Distributed Layer (TDL)+CNNs+LSTM; (2) vision transformers (ViT), both with pre-trained weights and trained from scratch; and (3) pre-trained video vision transformers (ViViT).

Consequently, our problem is most similar to the work of Baffa et al. [6]. Unlike their approach, we propose the use of pre-trained ViViT TimeSformer to address the sequential nature of DIT. Our results indicate that ViViTs exhibit performance comparable to 3D-CNNs for UFF's dynamic acquisition protocol in the DMR-IR dataset. However, there are important differences with respect to their work: (a) we work with a larger input tensor, providing higher resolution (their work resizes inputs to $(B, 20, 32, 32, 3)$, while we use $(B, 8, 224, 224, 3)$); (b) although the TimeSformer architecture reduces the number of frames processed, the model achieves similar performance.

Similar to Baffa et al., our unit of analysis is the patient, whereas Garia et al. focus on individual images since they focus on SIT protocol. Experiments using the basic ViT architecture show performance comparable to ViViT, specially when using 32×32 patches. But it is important to notice that ViT cannot process infrared (IR) images sequentially because they are inherently spatial and do not operate across the time dimension. As a result, frames are effectively averaged during processing.

Finally, our results suggest that further investigation is required when working directly with temperature matrices (TM). We also recommend that future research clearly indicate whether they are using IR images or raw temperature values.

Table 5: Performance comparison of different models on DIT data

Dataset	Model	Metric	Cross-Val (Mean ± Std)	Test Set
$\mathbb{D}_{\mathrm{TM}}$	TDL+CNN+LSTM	F1-score	0.8585 ± 0.0296 ($k = 3$)	-
		AUC	-	-
$\mathbb{D}_{\mathrm{IR}}$	ViT (16×16) pre-trained	F1-score	0.9644 ± 0.0458	-
		Accuracy	0.9538 ± 0.0615	0.87
		AUC	-	-
	ViT (32×32) pre-trained	F1-score	1.0000 ± 0.0000	0.9412
		Accuracy	1.0000 ± 0.0000	0.9333
		AUC	1.0000 ± 0.0000	0.9444
	ViT (32×32) scratch	F1-score	0.9101 ± 0.0665	0.8889
		Accuracy	0.8939 ± 0.0652	0.8667
		AUC	0.8866 ± 0.0595	0.8611
	ViViT	F1-score	0.9339 ± 0.0020	0.9412
		Accuracy	0.9231 ± 0.0000	0.9333
		AUC	0.9350 ± 0.0094	0.9444

Table 6: Comparison of studies on breast thermography classification

Study	Model	Protocol	Pre-trained	Dataset	Analysis Unit	Total Subjects	Cross Validation	Test Set	AUC
Garia et al [11]	ViT	SIT	No	DMR-IR	Images	950	-	10%	0.9570
Baffa et al [6]	3D-CNN	DIT	No	DMR-IR	Patient	149	K=10	-	0.9351
Our work	ViViT	DIT	Yes	DMR-IR	Patient	149	K=10	10%	**0.9444**

7 Discussion

To the best of our knowledge, this work presents the first application of a ViViT to the DIT protocol. Our approach achieves superior performance on a test set of 15 patients (93% accuracy, 93% precision, 94% recall) compared to the results reported in [6]. Furthermore, in a $K = 10$ cross-validation setting, our method achieves a similar mean accuracy of 93.50% but demonstrates superior stability, as indicated by a significantly smaller standard deviation of $\pm$ 0.94%. This higher precision and lower variance suggest a more robust model. It is noteworthy that our experiments utilize IR images with a spatial resolution of 224×224 pixels, a significantly larger input size compared to the 32×32 images used by the baseline method in [6]. We employ a shorter temporal sequence ($F = 8$ frames) to accommodate the use of a pre-trained model. To address this, future ablations studies should quantify the incremental value of temporal modeling by varying (a) the number of frames (e.g., 3, 4, 6, 8, and the full 20-frame sequence), (b) the sampling strategy (contiguous vs. uniformly spaced frames), and (c) the timing (early vs. late frames relative to cooling).

Another important difference is that our current results do not use a prior segmentation stage. Moreover, this arises the question if segmentation would increase performance in our results. However, as indicated in [20], there are no ablation studies that could confirm this in infrared TG.

Our experiments corroborate that ViTs with 32×32 patches yield better performance. However, it should be noted that a different acquisition protocol for the same dataset was employed compared to [11]. In contrast to their findings, pre-training clearly benefited performance in our experiments, despite the domain gap between natural images/videos and medical thermal data.

DIT aims to capture physiological responses under thermal stress that may not be visible in single static frames. Our results show that a pre-trained spatial ViT with temporal averaging can reach performance comparable to a pre-trained video transformer, suggesting that a substantial portion of the discriminative signal may be spatial. This raises two complementary questions: (i) whether the current DIT protocol sufficiently captures the temporal dynamics it is designed to measure, and (ii) whether simpler static protocols (SIT) augmented with robust spatial modeling could be adequate for certain clinical scenarios.

The results also suggest that future works should emphasize the type of thermal data used, as outcomes obtained directly from the point temperatures in CSV files differ from those using IR images. The challenges associated with temperature maps may stem from the fact that pre-trained networks are typically trained on natural images, which constitute a domain distinct from both IR imagery and temperature maps. We observed that infrared RGB images delivered stronger performance than raw temperature matrices.

7.1 Threats to Validity

Our study is primarily limited by: (i) the relatively small size of the dataset, and (ii) the absence of clinical validation. Although this research prevented patient information leakage in the hold-out test set, the limited sample size, combined with the use of large architectures such as transformers, increases the risk of overfitting. This risk, in turn, limits the potential for the clinical translation of these results. To mitigate overfitting, FT and data augmentation were employed; however, future work should include statistical tests and ablation studies to further validate and improve the model's performance. The sample size of available public datasets, however, is a factor over which we have limited control, besides some of the suggestions indicated in [20]. Clinical validation, while challenging to achieve, is necessary. It would not only validate the current model but also enhance it by incorporating other physiological factors (e.g., menstrual cycle, hormone therapy) that may influence thermographic patterns and, consequently, the model's performance.

8 Conclusions

This research paper shows that ViViTs can be effectively applied to BC classification using dynamically acquired TG images. Our current results show that the basic TimeSformer with 8 frames has achieved a similar performance to 3D-CNN in a prior work. Also, this work extends that of [11] by extending to the dynamic protocol.

With respect to the research questions, we conclude that transformer-based models exhibit a performance comparable to that of CNNs for this research problem. In our experiments, fine-tuning a pre-trained model yielded superior performance. Finally, the integration of DIT with ViViTs achieves results similar to those reported in [11]. Moreover, our results indicate that ViTs with temporal averaging perform similarly to fine-tuned ViViTs. This suggests that further research is necessary to fully account for the effect of temporal dynamics in BC prediction using infrared images and transformer-based architectures.

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References

1. Arnab, A., Dehghani, M., Heigold, G., Sun, C., Lučić, M., Schmid, C.: Vivit: A video vision transformer. Proceedings of the IEEE International Conference on Computer Vision pp. 6816–6826 (3 2021). https://doi.org/10.1109/ICCV48922.2021.00676, https://arxiv.org/pdf/2103.15691
2. Bai, S., Kolter, J.Z., Koltun, V.: An empirical evaluation of generic convolutional and recurrent networks for sequence modeling (3 2018), https://arxiv.org/pdf/1803.01271
3. Bertasius, G., Wang, H., Torresani, L.: Is space-time attention all you need for video understanding? Proceedings of Machine Learning Research **139**, 813–824 (2 2021), https://arxiv.org/pdf/2102.05095
4. Bray, F., Ferlay, J., Soerjomataram, I., Siegel, R.L., Torre, L.A., Jemal, A.: Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. CA: A Cancer Journal for Clinicians **68**(6), 394–424 (2018). https://doi.org/10.3322/caac.21492, http://doi.wiley.com/10.3322/caac.21492
5. Chollet, F., et al.: Keras. https://keras.io (2015)
6. De, M., Baffa, F.O., Lattari, L.G., Conci, A.: 3d convolutional neural networks for dynamic breast infrared imaging classification. Tech. rep. (2023)
7. Dhage, J.S., Gulve, A.K.: Time-distributed layer convolutions with long short-term memory for human activity recognition and result comparison with various machine learning models. Engineering, Technology and Applied Science Research **15**, 23277–23282 (6 2025). https://doi.org/10.48084/etasr.10499
8. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al.: An image is worth 16x16 words. arXiv preprint arXiv:2010.11929 **7** (2020)
9. Falconí Estrada, L.G.: Transfer and ensemble learning models in breast mammogram pathology classification. Master's thesis, Escuela Politécnica Nacional, Quito, Ecuador (2020), https://bibdigital.epn.edu.ec/handle/15000/20853
10. Fan, H., Xiong, B., Mangalam, K., Li, Y., Yan, Z., Malik, J., Feichtenhofer, C.: Multiscale vision transformers. Proceedings of the IEEE International Conference on Computer Vision pp. 6804–6815 (4 2021). https://doi.org/10.1109/ICCV48922.2021.00675, https://arxiv.org/pdf/2104.11227
11. Garia, L.S., Hariharan, M.: Vision transformers for breast cancer classification from thermal images. Lecture Notes in Electrical Engineering **1009 LNEE**, 177–185 (2023). https://doi.org/10.1007/978-981-99-0236-1_13, https://link.springer.com/chapter/10.1007/978-981-99-0236-1_13
12. Ji, S., Xu, W., Yang, M., Yu, K.: 3d convolutional neural networks for human action recognition. IEEE Transactions on Pattern Analysis and Machine Intelligence **35**, 221–231 (2013). https://doi.org/10.1109/TPAMI.2012.59
13. Kennedy, D.A., Lee, T., Seely, D.: A comparative review of thermography as a breast cancer screening technique (3 2009). https://doi.org/10.1177/1534735408326171
14. Kitchenham, B., Brereton, O.P., Budgen, D., Turner, M., Bailey, J., Linkman, S.: Systematic literature reviews in software engineering–a systematic literature review. Information and software technology **51**(1), 7–15 (2009)
15. Kolesnikov, A., Beyer, L., Zhai, X., Puigcerver, J., Yung, J., Gelly, S., Houlsby, N.: Big transfer (bit): General visual representation learning. Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and

Lecture Notes in Bioinformatics) **12350 LNCS**, 491–507 (12 2019). https://doi.org/10.1007/978-3-030-58558-7_29, https://arxiv.org/pdf/1912.11370

16. Li, Y., Liang, Q., Gan, B., Cui, X.: Action recognition and detection based on deep learning: A comprehensive summary. Computers, Materials and Continua **77**, 1–23 (10 2023). https://doi.org/10.32604/CMC.2023.042494, https://www.sciencedirect.com/org/science/article/pii/S1546221823001285
17. Litjens, G., Kooi, T., Bejnordi, B.E., Setio, A.A.A., Ciompi, F., Ghafoorian, M., van der Laak, J.A., van Ginneken, B., Sánchez, C.I.: A survey on deep learning in medical image analysis. Medical Image Analysis **42**, 60–88 (12 2017). https://doi.org/10.1016/J.MEDIA.2017.07.005
18. Liu, D., Wu, B., Li, C., Sun, Z., Zhang, N.: TrenD: A transformer-based encoder-decoder model with adaptive patch embedding for mass segmentation in mammograms. Medical Physics **50**, 2884–2899 (5 2023). https://doi.org/10.1002/mp.16216
19. Rakhunde, M.B., Gotarkar, S., Choudhari, S.G.: Thermography as a breast cancer screening technique: A review article. Cureus (11 2022). https://doi.org/10.7759/cureus.31251
20. Ryan, L., Agaian, S.: Breast cancer detection using infrared thermography: A survey of texture analysis and machine learning approaches. Bioengineering **12**, 639 (6 2025). https://doi.org/10.3390/BIOENGINEERING12060639, https://pmc.ncbi.nlm.nih.gov/articles/PMC12189745/
21. Selva, J., Johansen, A.S., Escalera, S., Nasrollahi, K., Moeslund, T.B., Clapés, A.: Video transformers: A survey. IEEE Transactions on Pattern Analysis and Machine Intelligence **45**(11), 12922–12943 (Nov 2023). https://doi.org/10.1109/tpami.2023.3243465, http://dx.doi.org/10.1109/TPAMI.2023.3243465
22. Silva, L.F., Saade, D.C., Sequeiros, G.O., Silva, A.C., Paiva, A.C., Bravo, R.S., Conci, A.: A new database for breast research with infrared image. Journal of Medical Imaging and Health Informatics **4**, 92–100 (2014). https://doi.org/10.1166/jmihi.2014.1226
23. Su, Y., Liu, Q., Xie, W., Hu, P.: YOLO-LOGO: A transformer-based YOLO segmentation model for breast mass detection and segmentation in digital mammograms. Computer Methods and Programs in Biomedicine **221** (6 2022). https://doi.org/10.1016/j.cmpb.2022.106903
24. Tran, D., Wang, H., Torresani, L., Ray, J., Lecun, Y., Paluri, M.: A closer look at spatiotemporal convolutions for action recognition. Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition pp. 6450–6459 (11 2017). https://doi.org/10.1109/CVPR.2018.00675, https://arxiv.org/pdf/1711.11248
25. Tsietso, D., Yahya, A., Samikannu, R.: A review on thermal imaging-based breast cancer detection using deep learning. Mobile Information Systems **2022**(1), 8952849 (2022)
26. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Kaiser, L.u., Polosukhin, I.: Attention is all you need. In: Guyon, I., Luxburg, U.V., Bengio, S., Wallach, H., Fergus, R., Vishwanathan, S., Garnett, R. (eds.) Advances in Neural Information Processing Systems. vol. 30. Curran Associates, Inc. (2017), https://proceedings.neurips.cc/paper_files/paper/2017/file/3f5ee243547dee91fbd053c1c4a845aa-Paper.pdf
27. Visual-Lab: Captura de imagens térmicas no huap (rotina diária e protocolos) (2014), https://visual.ic.uff.br/proeng/protocolo.pdf